Ignacio **Peis**

Technical University of Denmark, DTU
Dept. of Applied Mathematics and Computer Science
Section for Cognitive Systems
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Google Scholar

Positions

Technical University of Denmark, DTU — Oct. 2023 - Present
Postdoctoral Researcher
Section for Cognitive Systems
Dept. of Applied Mathematics and Computer Science
(DTU Compute)

Intor.AI — Nov. 2024 - Present
AI Researcher

EBIS Business School — Oct. 2024 - Present
Lecturer

Fundación UC3M & BBVA — Dec. 2020 - Dec. 2024
Mentor at specialized courses

Universidad Carlos III de Madrid — Oct. 2019 - Sept. 2023
Predoctoral Researcher & Teaching Assistant
Signal Processing Group
Supervisors: Prof. Antonio Artés Rodríguez
Dr. Pablo Martínez Olmos

University of Cambridge — Feb. 2021 - Feb. 2022
Visiting researcher
Machine Learning Group
Supervisor: Prof. José Miguel Hernández-Lobato

Universidad Carlos III de Madrid — Feb. 2017 - Sept. 2019
Research associate
Signal Processing Group
Supervisor: Prof. Antonio Artés Rodríguez

Affiliations

"La Caixa" Foundation — Mar 2026 - Present
Postdoctoral Junior Leader Incoming

Pioneer Centre for Artificial Intelligence — Nov 2023 - Present
Postdoctoral Researcher

Danish Data Science Academy (DDSA) — June 2023 - Present
Postdoc Fellow

Education

Universidad Carlos III de Madrid — Oct 2018 - Oct 2023
PhD in Probabilistic Machine Learning
Advisors: Prof. Antonio Artés Rodríguez
Dr. Pablo Martínez Olmos
Cum Laude
Outstanding PhD Thesis Award

Universidad Carlos III de Madrid — Sep. 2016 - Jul. 2018
M.Sc. in Telecommunications Engineering
Dissertation with highest mark

Universidad Carlos III de Madrid — Sep. 2016 - Jul. 2018
M.Sc. in Multimedia and Communications
Dissertation with highest mark
One course with honors

Universidad de Granada — Sep. 2012 - Jul. 2016
B.Sc. in Telecommunications Engineering
Five courses with honors

Grants

Junior Leader Incoming Postdoctoral Fellowship

Title:	*"Flow-Based Generative Meta-Learning for Reliable AI in Medicine"*
Role:	Principal Investigator
Success rate:	5.93%
Budget:	320.000,00 €
Period:	2026-2029
Entity:	*La Caixa Foundation*
Country:	Spain

DDSA Postdoctoral Fellowship

Title:	*"Implicit Neural Representations Generative Modelling"*
Role:	Principal Investigator
Success rate:	11.32%
Budget:	DKK 1.200.000,00 (approx. 160.597,20 €)
Period:	2023-2025
Entity:	Danish Data Science Academy (DDSA) Novo Nordisk Foundation
Country:	Denmark

FPU Doctoral Fellowsip

Role:	PhD Student
Success rate:	
Period:	2019-2023
Entity:	Ministry of Science, Innovation and Universities
Country:	Spain

Awards

Best Reviewer Award — 2025
International Conference on Artificial Intelligence and Statistics (**AISTATS**)

Top Reviewer 2024
Neural Information Processing Systems (**NeurIPS**)

Top-10% Reviewer 2023
International Conference on Artificial Intelligence and Statistics (**AISTATS**)

Outstanding PhD Thesis Award 2023
Universidad Carlos III de Madrid
PhD Defense Committee:
Dr. Daniel Hernández Lobato, UAM
Dr. Francisco R. Ruiz, Google Deepmind
Prof. Fernando Pérez Cruz, ETH, SDSC

Conference Papers

Ignacio Peis, Batuhan Koyuncu, Isabel Valera and Jes Frellsen. Hyper-Transforming Latent Diffusion Models. *Proceedings of the 42nd* ***International Conference on Machine Learning****, in Proceedings of Machine Learning Research*, 2025.
[pdf] [poster] [video] [slides] [code]

Batuhan Koyuncu, Pablo Sánchez-Martín, **Ignacio Peis**, Pablo M. Olmos and Isabel Valera. Variational Mixture of HyperGenerators for Learning Distributions Over Functions. *Proceedings of the 40th* ***International Conference on Machine Learning****, in Proceedings of Machine Learning Research 202:17660-17683*, 2023.
[pdf] [poster] [video] [code]

Ignacio Peis, Chao Ma and José Miguel Hernández-Lobato. Missing Data Imputation and Acquisition with Deep Hierarchical Models and Hamiltonian Monte Carlo. In *Advances in Neural Information Processing Systems (****NeurIPS****) 35: 35839-35851*, 2022.
[pdf] [code] [slides] [video] [poster]

Ignacio Peis, Ignacio A. Illán, Francisco J.Martínez-Murcia, Fermín Segovia, Juan M. Górriz, Javier Ramírez, E. W. Lang & Diego Salas-Gonzalez. MRI brain segmentation using hidden Markov random fields with alpha-stable distributions. In *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room-Temperature Semiconductor Detector Workshop (****NSS/MIC/RTSD****)* (pp. 1-3). IEEE, 2016.
[html]

Journal Papers

Berian James, Stefan Pollok, **Ignacio Peis**, Jes Frellsen and Rasmus Bjørk. Scalable physical source-to-field inference with hypernetworks. ***Transactions of Machine Learning Research****, 2835-8856*, 2026.
[pdf] [code]

Ignacio Peis, Pablo M. Olmos and Antonio Artés-Rodríguez. Unsupervised Learning of Global Factors in Deep Generative Models. In ***Pattern Recognition****, vol. 134, p. 109130*, 2022.
[pdf] [code]

Ignacio Peis, Javier-David López-Moríñigo, M. Mercedes Pérez-Rodríguez, Maria-Luisa Barrigón, Marta Ruiz-Gómez, Antonio Artés-Rodríguez and Enrique Baca-García. Actigraphic recording of motor activity in depressed inpatients: a novel computational approach to prediction of clinical course and hospital discharge. In ***Scientific reports**, 10.* Nature, 2020.
[pdf]

Ignacio Peis, Pablo M. Olmos, Constanza Vera-Varela, Maria-Luisa Barrigón, Philippe Courtet, Enrique Baca-García and Antonio Artés-Rodríguez. Deep Sequential Models for Suicidal Ideation from Multiple Source Data. In ***Journal of Biomedical and Health Informatics**, vol. 23, no. 6.* IEEE, 2019.
[pdf] [html]

D. Castillo-Barnes, **I. Peis**, F. J. Martínez-Murcia, F. Segovia, I. A. Illán, J. M. Górriz, J. Ramírez, D. Salas-Gonzalez. **A Heavy Tailed Expectation Maximization Hidden Markov Random Field Model with Applications to Segmentation of MRI**. In *Frontiers in Neuroinformatics, 11, 66*, 2017.
[pdf]

Teaching

Data Mining, Machine Learning and Deep Learning — 2024 - Present
M.Sc. in Data Science and Artificial Intelligence
EBIS, **EUNEIZ**, **Microsoft**

Generative AI — 2025 - Present
Specialized Course
BBVA, **Fundación UC3M**

Natural Language Processing — 2026 - Present
Specialized Course
BBVA, **Fundación UC3M**

Advanced Machine Learning — 2023 - Present
Specialized Course
BBVA, **Fundación UC3M**

Intermediate Machine Learning and Feature Engineering — 2022 - Present
Specialized Course
BBVA, **Fundación UC3M**

Machine Learning Fundamentals — 2020 - Present
BBVA, **Fundación UC3M**

Machine Learning II — 2020, 2022
Bachelor in Data Science and Engineering
Teaching evaluation survey: 4.67/5 (2020), 4.63/5 (2022)
Universidad Carlos III de Madrid

Neural Networks — 2022
Bachelor in Data Science and Engineering
Teaching evaluation survey: 4.13/5

Universidad Carlos III de Madrid

Communications Theory 2022
Bachelor in Telecommunications Engineering
Bachelor in Sound and Image Engineering
Teaching evaluation survey: 3.94/5, 4.14/5
Universidad Carlos III de Madrid

Supervision

Master Students at Technical University of Denmark 2025 - 2026
Noa Margeta (2025), Nikolaj Bach Meineche (2025), Eva Kastelan (2026), Frederik Emil Nagel (2026)

Visting PhD Students at Technical University of Denmark 2026
Francisco Javier Sáez Maldonado (Universidad de Granada, 2026)

Service

Generative Modelling Summer School (GeMMS) 2027
Local Organizer

Andaluz.IA Conference 2025 - Present
Scientific Commitee

Talks

Hyper-Transforming Latent Diffusion Models Dec. 2025
Andaluz.IA forum
[slides]

Hyper-Transforming Latent Diffusion Models Dec. 2025
Signal Processing Group
Universidad Carlos III de Madrid

Hyper-Transforming Latent Diffusion Models Jun. 2025
Section for Cognitive Systems
[slides]

Hyper-Transforming Latent Diffusion Models Apr. 2025
CITIC, Universidad de Granada
[slides]

Variational Mixture of HyperGenerators for Learning Distributions Over Functions Dec. 2023
Andaluz.IA forum
[slides]

Information Acquisition and Distributions of Functions with Deep Generative Models Jun. 2023
Pioneer Centre for Artificial Intelligence, Copenhagen, Denmark
[slides]

Missing Data Imputation and Acquisition with Deep Hierarchical Models and Hamiltonian Monte Carlo — Dec. 2022
Oral (video) and poster presentation
NeurIPS22, New Orleans, USA
[slides] [video] [poster]

Missing Data Imputation and Acquisition with Deep Hierarchical Models and Hamiltonian Monte Carlo — Jun. 2022
Signal Processing Group
Universidad Carlos III de Madrid

Deep Sequential Models for Suicidal Ideation from Multiple Source Data — Jul. 2018
Signal Processing Group
Universidad Carlos III de Madrid

Reviewing

Pattern Recognition, Elsevier	2026 - Present
The European Conference on Artificial Intelligence (**ECAI**)	2025 - Present
Workshop on Weight-Space Learning (**ICLR**)	2025 - Present
International Conference on Machine Learning (**ICML**)	2025 - Present
Transactions on Machine Learning Research (**TMLR**)	2025 - Present
Northern Lights Deep Learning Conference (**NLDL**)	2025 - Present
Workshop Reviewer (**EurIPS**)	2025
International Conference on Learning Representations (**ICLR**)	2024 - Present
Neural Information Processing Systems (**NeurIPS**)	2024 - Present
Top Reviewer (2024)	
Association for Uncertainty in Artificial Intelligence (**UAI**)	2024 - Present
Artificial Intelligence and Statistics (**AISTATS**)	2022 - Present
Top-10% Reviewer (2023)	
Best Reviewer Award (2025)	
Journal of Machine Learning Research (**JMLR**)	2022 - Present
Journal of Biomedical and Health Informatics (**JBHI**)	2018 - Present
Workshop on Deep Generative Models for Health (**DGM4H**)	2023

Projects

Title: **PRACTICO-CM: *"Psiquiatría Computacional y Modelos Integrales de Comportamiento"***
Role: Research team member
Budget: 645.775,90 €
Period: 2019 - 2022
Entity: Consejería de Educación e Investigación, CAM
Country: Spain

Title: ***"Aprendizaje máquina y computación masiva para medicina personalizada y análisis cuantitativo del clima"***
Role: Research team member
Budget: 124.509,00 €
Period: 2019 - 2022
Entity: Agencia Estatal de Investigación (AEI)
Country: Spain

Title:	***"Métodos computacionales bayesianos avanzados pare estimación, predicción y control en sistemas multisensoriales complejos"***
Role:	Research team member
Budget:	314.600,00 €
Period:	2016 - 2029
Entity:	Ministerio de Economía, Comercio y Empresa
Country:	Spain

Title:	***"Modelos profundos y explicables basados en variables latentes para salud mental"***
Role:	Research team member
Budget:	190.212,00 €
Period:	2022 - 2025
Entity:	Agencia Estatal de Investigación (AEI)
Country:	Spain

Courses

Model Parallelism: Building and Deploying Large Neural Networks — Oct. 2024
NVIDIA

AI and Machine Learning in Healthcare Summer School — Sep. 2022
Cambridge Centre for AI in Medicine
University of Cambridge, UK

Gaussian Process and Uncertainty Quantification Summer School, 2020. — Sep. 2020
University of Sheffield, UK

Machine Learning Summer School (MLSS) — Sep. 2019
Skoltech Institute of Science and Technology, Moscow, Russia

Machine Learning Frontiers in Precision Medicine (MLFPM) 1st Summer School — Sep. 2019
ETH Zurich, Switzerland

Machine Learning Summer School (MLSS) — Sep. 2018
Universidad Autónoma de Madrid, Spain

Gaussian Process and Uncertainty Quantification Summer School, 2017. — Sep. 2017
University of Sheffield, UK

Dissertations

Ignacio Peis. Advanced Inference and Representation Learning Methods in Variational Autoencoders. ***PhD Thesis Dissertation*** *(Probabilistic Machine Learning), 2023.* [pdf] [slides]

Ignacio Peis. Deep sequential models with attention for psychiatric patients clinical assessment. ***M.Sc. Thesis Dissertation*** *(Multimedia and Communications), 2018.*

Ignacio Peis. Activity monitoring in depressed patients in the hospital setting: a pilot study testing new methods of actigraphy data analysis for predicting clinical progress and date of hospital discharge. ***M.Sc. Thesis Dissertation*** *(Telecommunications Engineering), 2018.*

Ignacio Peis. Hidden Markov Random Fields with alpha-stable distributions for brain Magnetic Resonance Images. ***B.Sc. Thesis Dissertation*** *(Telecommunications Engineering), 2016.*

Last updated: March 30, 2026